Carlos Andrés Montoya Tobón



Java Solution Architect & Tech Lead

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CORE STACK Java · Spring Boot · Microservices · AWS · Hexagonal Architecture · Keycloak · PostgreSQL · RabbitMQ

PROFESSIONAL SUMMARY

I'm a Java Solution Architect and Tech Lead based in Medellín, Colombia, with 18+ years in backend engineering: 11+ of them leading teams of up to 45 people, and nearly 7 as a solutions architect. I've designed and built Java platforms ranging from transportation management systems and telecom self-service portals to fintech payments, digital identity and blockchain-verified ticketing. My specialty is backend architecture on the Spring ecosystem: hexagonal microservices and event-driven systems on Kafka and RabbitMQ, built for resilience and idempotency, secured with Keycloak and observable end to end, deployed on AWS, GCP and Azure. As a lead I set the architecture, grow the team technically and own delivery, backed by DevOps culture and agile practice (certified Scrum Master).

EXPERIENCE

Solution Architect & Tech Lead

May 2026 — Present

Soldife · Remote

- Lead an 8-person team (3 frontend, 3 backend and 2 QA) as Tech Lead: I set the architecture, split and review the work, and coach each profile, owning the product end to end, from the Spring Boot backend to the React frontend.
- Architected Mr. Yoker, a ticketing platform where every ticket is a W3C Verifiable Credential, cryptographically signed and anchored on blockchain: tickets can't be forged or duplicated, and anyone can verify one without calling the issuer.
- Own platform security: credential cryptography, centralized identity at the gateway with Keycloak and mandatory OWASP scanning in every pipeline.
- Set the architectural standard (hexagonal + DDD) across 6 microservices and 9 auto-configurable Spring Boot SDKs, with contract governance (SemVer, ADRs) and boundaries enforced by ArchUnit on every build: zero architecture drift between PRs.
- Designed asynchronous messaging on RabbitMQ with key resilience and idempotency patterns (transactional outbox, canonical event envelope, idempotent processing), so no event is lost or processed twice.
- Built the platform's observability: end-to-end W3C tracing and multiple Grafana dashboards for running the services.
- Automated delivery: GitHub Actions CI/CD with reusable workflows and quality gates (tests, coverage, OWASP, ArchUnit, Spotless), plus versioned GHCR base images (Temurin 21, GraalVM) that speed up pipelines.
- Wrote architecture guidelines executable by AI coding agents (Claude Code) and documented the architecture as code (C4 in Structurizr, ADRs), so the team ships faster without breaking the standard.

Java · Spring Boot · React · PostgreSQL · Keycloak · RabbitMQ · Blockchain · W3C Verifiable Credentials · Cryptography · Hexagonal Architecture · DDD · Microservices · Grafana · KrakenD · ArchUnit · GitHub Actions · Claude Code

Backend Developer

Sep 2024 — Jan 2026

F2X - Flypass · Medellín, Colombia

- Designed and built membership and commission billing for every Flypass service, and resolved critical production incidents in payment transactionality with Bancolombia by applying resilience and idempotency patterns on Kafka that eliminate duplicate charges (Java · Spring Boot · hexagonal architecture on AWS · PostgreSQL).
- Designed and built the fuel payment flow, a new product inside the Flypass app, from design through production readiness.
- Built multiple Grafana dashboards to monitor the billing and payment flows, and owned incident response and operational support for the backend services.

Java · Spring Boot · Kafka · PostgreSQL · AWS · Hexagonal Architecture · Microservices · Grafana

Backend Developer

Dec 2023 — Dec 2024

Blerify · Remote

- Built API security for Blerify, a digital-identity startup based on verifiable credentials operating across LATAM: multi-organization authentication and authorization with Keycloak, with roles and permissions isolated per organization.
- Implemented KrakenD as the platform's API gateway: a single entry point that validates tokens at the edge before requests reach the services.
- Designed and built the transactional module for memberships and service billing.
- Developed the blockchain-based voucher/bond issuance component (Java · Vert.x · Kotlin · PostgreSQL on GCP/Kubernetes).

Java · Vert.x · Kotlin · Keycloak · KrakenD · Blockchain · PostgreSQL · GCP · Kubernetes

Backend Developer
Apr 2022 — Jun 2023

Tul · Medellín, Colombia

- Built the warehouse and inventory management modules for Tul's B2C marketplace (an end-user app), on Java/Spring Boot hexagonal microservices on AWS and PostgreSQL.
- Integrated the services through Kafka events, applying resilience patterns to inter-service communication.
- Implemented service security with Spring Security and observability with CloudWatch and Mezmo (centralized logging) for running them in production.
- Ran platform-wide code-quality reviews.

Java · Spring Boot · Kafka · PostgreSQL · AWS · Spring Security · CloudWatch · Mezmo · Hexagonal Architecture · Microservices

Solution Architect & Tech Lead
Aug 2020 — Apr 2022

k4s - Klever for Solutions · Medellín, Colombia

- As Solution Architect and Tech Lead, led a cross-functional team of 8 (frontend, backend, QA and PO) through the platform transformation of MutualSER, a health insurer (EPS) based in Cartagena, from architecture to delivery.
- Defined the new platform's PaaS architecture on AWS and the reusable security and Keycloak/GCP integration frameworks consumed by several product teams.
- Developed the Java/Spring Boot backend and a micro-frontend front end (React + Module Federation) secured with Keycloak.
- Grew the team's technical level through mentoring in Spring, SOLID principles and clean architecture.

Java · Spring Boot · React · Module Federation · Keycloak · AWS · GCP · Microservices · Clean Architecture

Solution Architect & Tech Lead
May 2016 — Apr 2020

Integral Software SAS · Medellín, Colombia

- As Solution Architect and Tech Lead, led a cross-functional team of 4 (backend and frontend) building products for the City of Medellín, Universidad de Medellín, Indumil and the Colombian Air Force (FAC).
- Designed architectures and systems for the City of Medellín — among them Atenea and Tienda Virtual — with hexagonal architecture, design patterns and security, including their traceability and accountability reporting layer.
- Built an internal portal for Indumil (Colombia's state defense manufacturer) that generates payment receipts through SOAP integration with its systems.
- Led the migration of legacy systems to Spring Boot with hexagonal architecture for Universidad de Medellín and the FAC, on PostgreSQL, Oracle and SQL Server, integrating the existing SOAP services with resilience patterns and Grafana observability.
- Built React and Angular SPAs on Spring backends (Java 8/11 and Kotlin), with authentication integrated with Keycloak.

Java · Spring Boot · Hexagonal Architecture · PostgreSQL · Oracle SQL · SQL Server · SOAP · Grafana · Keycloak · React · Angular · Kotlin

Solution Architect & Tech Lead
Sep 2015 — May 2016

Software Estratégico · Medellín, Colombia

- As Solution Architect and Tech Lead, led a 6-person team modernizing Matrix Tech's product line: new architectures on Spring Boot, RabbitMQ and Cassandra, integrated with the existing Oracle databases and SOAP services.
- Built an SMS messaging gateway from scratch.
- Led the adoption of a DevOps culture across IT areas which, together with the new Spring architecture, left teams delivering faster, more often and more reliably.

Java · Spring Boot · RabbitMQ · Cassandra · Oracle SQL · SOAP · DevOps

Technical Lead
Aug 2013 — Sep 2015

Tata Consultancy Services · Medellín, Colombia

- Led a distributed team of 45+ developers building Claro Perú's transactional self-service portal, from design through to its production release.
- Designed and built the Oracle Service Bus integration layer connecting Claro Colombia's back-office systems.
- Built the portal itself on Java 7/8, Spring and WebSphere Portlets, front end included, against IBM WebSphere.
- Stood up the continuous-integration infrastructure and introduced Scrum, giving a 45-person team a single repeatable delivery cadence.

Java · Spring · WebSphere Portlets · Oracle Service Bus · Jenkins

Tech Lead
Nov 2010 — Aug 2013

PATH S.A. · Medellín, Colombia

- As Tech Lead, designed the architectures of multiple company products: geolocation and vehicle tracking, a B2C SMS platform with business intelligence, notification systems (bulk email and SMS) and IoT pilots for the city of Medellín.
- Built the reporting system on Nokia NetPlanet for analyzing Claro's network.
- Designed the AWS and on-premise infrastructure and the CI/CD pipelines for the whole organization.

Java · AWS · Geolocation · SMS · IoT · Business Intelligence · CI/CD

Java Web Developer
Jun 2009 — Jun 2010

Ceiba Software House · Medellín, Colombia

- Built the loan, policy and occupational-risk systems of Suramericana — Colombia's largest insurer — on Java / JEE 5 with Struts and JSF.
- Wrote the Oracle data layer behind those flows: SQL and stored procedures carrying the policy and claims operations.
- Delivered every slice full-stack — Java back end, JavaScript / HTML / CSS front end and Oracle schema — deployed on Tomcat.

Java · JEE 5 · Struts · JSF · JSP · SOAP · Oracle SQL · JavaScript · HTML · Tomcat

Java Web Developer Apr 2007 — Jun 2009

Castor SAS (antes Mercurio) · Medellín, Colombia

- Designed and built the business components of a Transportation Management System that ran the daily operation of 25+ transport companies.
- Built the integration layer that carried TMS operations into each client's accounting system, removing the manual re-keying between both worlds.
- Owned support and maintenance for every company on the platform — production incidents, SQL Server / MySQL data fixes and JSP + JavaScript front-end work.
- Introduced Scrum and extreme-programming practices to a team that until then delivered ad hoc.

Java · Servlets · JSP · SOAP · JavaScript · HTML · Oracle SQL · SQL Server · MySQL · Tomcat

LIVE SITES

MyPipelines **mypipelines.andresmontoyat.dev** — Reusable CI/CD pipeline library — parameterized GitHub Actions workflows (build, tag-filtered tests, JaCoCo coverage, OWASP scan, ArchUnit, Spotless) shared across JVM, KrakenD and NGINX projects.

MyGuidelines — Engineering Standards **myguidelines.andresmontoyat.dev** — My engineering playbook, codified — the architecture, testing, security and clean-code standards I hold every project to, and the trade-offs behind them. Hexagonal + DDD for JVM microservices, written to be executable by engineers and AI coding agents.

TECHNICAL SKILLS

Backend	**Java** · **Kotlin** · Spring Boot · Spring Cloud · Spring Security · Hibernate · Kafka · RabbitMQ · Cassandra · REST / gRPC · Keycloak
Cloud & Infrastructure	**AWS** · GCP · Azure · **Kubernetes** · Docker · GKE · KrakenD · Terraform
DevOps & Tools	Jenkins · GitHub Actions · SonarQube · Nexus · Gradle · **Maven** · PostgreSQL · MySQL · Oracle SQL · Scrum Master
AI & Productivity	**Claude Code** · GitHub Copilot · JetBrains Junie · Prompt Engineering